

Table Tennis League Analysis

Portfolio Project Description

Project Background

As an active participant in a local table tennis league, I wanted to go beyond the limited statistics available on the official competition website. While the website publishes standings and match results after each round, it does not provide the detailed performance analytics needed to evaluate player development, compare teams, or prepare strategically for upcoming matches.

To address this challenge, I designed and developed a complete data pipeline that automatically collects, processes, and visualizes league data. The project combines Python-based web scraping, data transformation, anonymization, and Tableau dashboard development to create a practical sports analytics solution that is used throughout the season.

Project Objectives

- 1 Automate weekly collection of tournament data.
- 2 Create structured datasets from raw HTML pages.
- 3 Track team and player performance over time.
- 4 Support match preparation through data-driven insights.
- 5 Build professional Tableau dashboards for interactive analysis.

Data Collection and Engineering

The source data originates from the league website, where information is published after every round. Using Python, Requests, and BeautifulSoup, I built a modular scraping application capable of downloading, parsing, and extracting information from multiple page types including team standings, player rankings, match results, and individual player encounters.

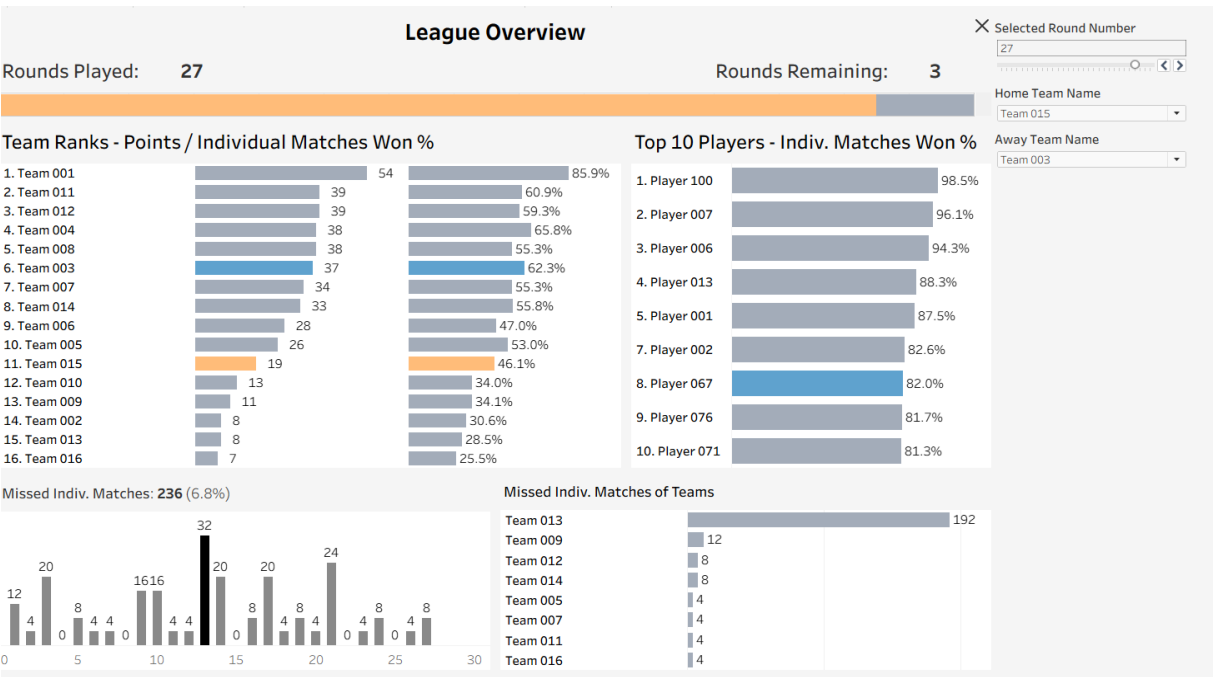
During development, AI-assisted exploration was used to better understand the structure of the website and accelerate implementation of the extraction logic. The resulting application follows an ETL-style workflow consisting of data acquisition, transformation, normalization, and anonymization.

Analytics and Visualization

After processing the data, the final datasets are loaded into Tableau where they are transformed into interactive dashboards. Two primary dashboards were developed.

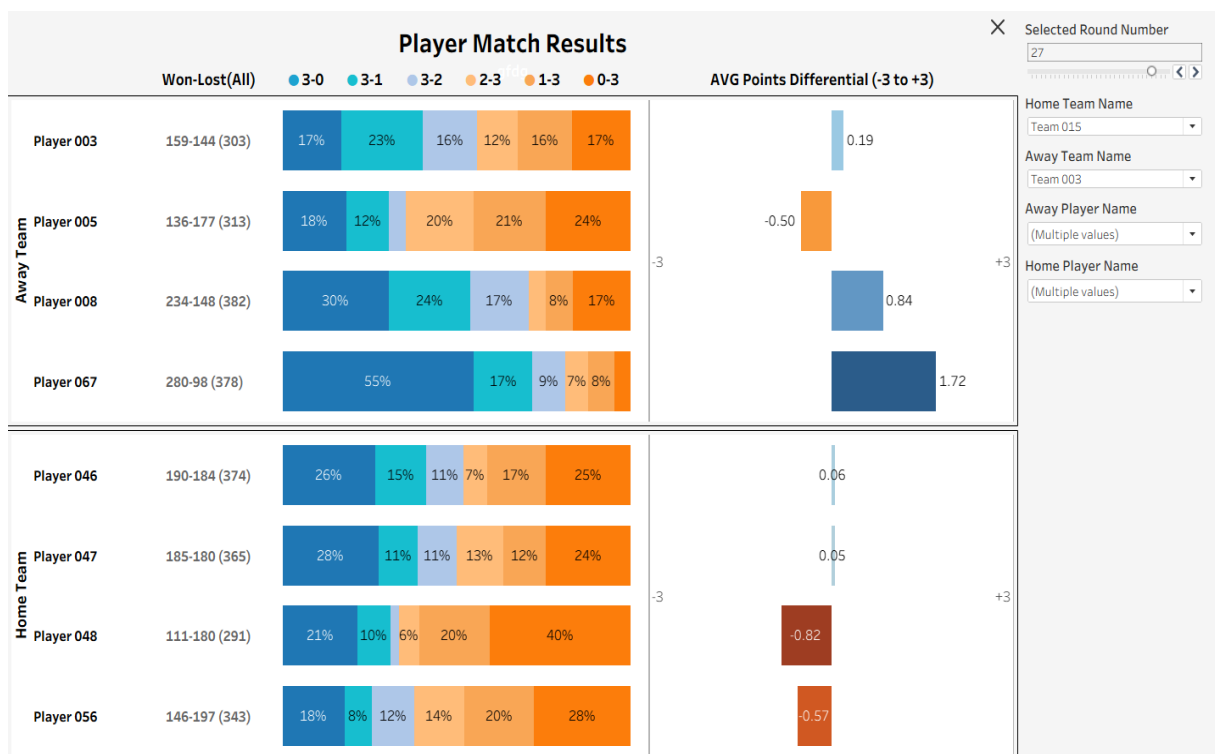
1. League Overview Dashboard

This dashboard provides a high-level view of the entire competition. It displays current standings, team rankings, points, individual match win percentages, tournament progress, participation statistics, and the highest-performing players. Users can filter the data by round and team to monitor changes throughout the season.



2. Player Performance Dashboard

The second dashboard focuses on individual player analysis. It enables comparison between players from opposing teams by examining win-loss records, match result distributions, and average point differentials. This helps identify strengths, weaknesses, consistency trends, and potential matchup advantages before future fixtures.



Business Value and Key Outcomes

- 1 Automated weekly data collection and processing.
- 2 Reusable ETL pipeline for sports analytics.
- 3 Improved visibility into team and player performance.
- 4 Interactive dashboards supporting tactical decisions.
- 5 Practical demonstration of analytics and data engineering skills.

Technologies Used

Python • Requests • BeautifulSoup • CSV Processing • Data Cleaning • Data Transformation • Data Anonymization • Tableau • Dashboard Design • Data Visualization • ETL Pipelines • PyCharm • AI-Assisted Development

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